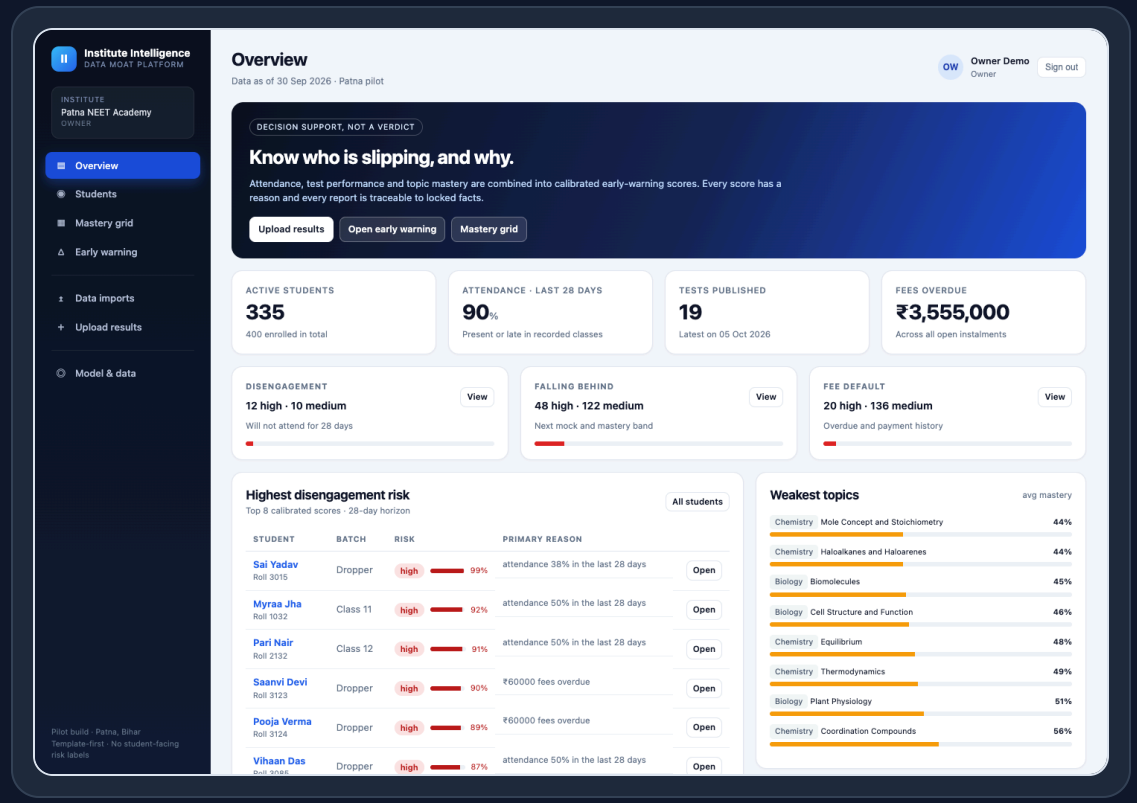
CASE B · THE DATA MOAT

The data is already there. The warning is not.

A privacy-first analytics layer that turns scattered test, attendance and fee data into timely action for Tier-2 coaching institutes.

NEET-first · Patna | Rasch + calibrated risk | DPDP-ready pilot



Live prototype · Patna NEET Academy
1,000 students · 181k responses · 3 institutes

THE PROBLEM

Coaching institutes are flying blind at the moment it matters.

The signals exist. They are trapped in OMR sheets, Excel files and WhatsApp threads.

01 · SIGNALS ARE FRAGMENTED

Tests, attendance and fees live in three to five disconnected tools. No one joins them at the student level.

02 · ACTION ARRIVES LATE

Owners see dropout after attendance collapses. Teachers cannot see which topic needs reteaching this week.

03 · PARENTS GET A RANK

A percentile without a reason or a next step creates anxiety, not trust — and does not retain the student.

The moat is not more content. It is the workflow that turns existing data into timely, explainable action.

TODAY

Monthly Excel review · rank cards · ad-hoc WhatsApp · no shared reason code

WITH INSTITUTE INTELLIGENCE

Weekly 20-minute review · named reasons · parent update · audited action

THE PRODUCT LOOP

One weekly loop: measure, explain, act.

Every score is traceable to a feature. Every parent report is traceable to locked facts.

1

INGEST

CSV / Forms export

Validate, quarantine, commit. Idempotent jobs.

2

MEASURE

Rasch + topic mastery

Shrinkage, recency, anchor items.

3

EXPLAIN

Risks with reasons

Calibrated disengagement; transparent rules.

4

ACT

Intervention + report

Teacher action, EN/HI report, WhatsApp queue.

28 days
disengagement horizon

3
risk targets, separate thresholds

2
languages, one locked facts object

Every weekly loop creates labelled outcomes: better calibration, better benchmarks, higher switching cost.

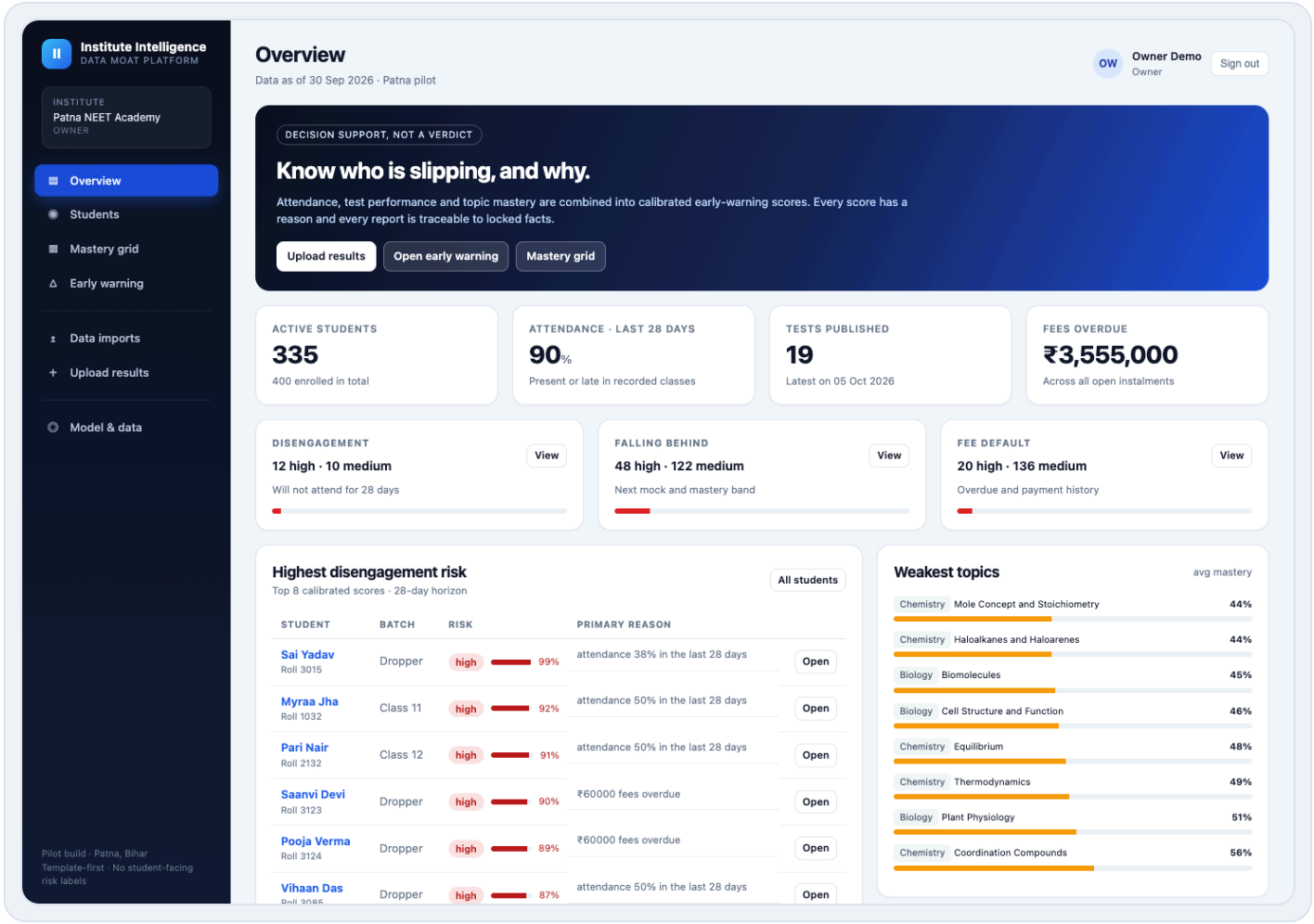
THE PRODUCT

Know who is slipping, and why.

A daily command centre for the owner, with reasons a teacher can act on.

- **Three targets, three thresholds**
  Disengagement, falling behind, fee default. Fee data is redacted for teachers.
- **Rasch, not raw percentages**
  Topic mastery with uncertainty and low-evidence labels.
- **From flag to action**
  Interventions are teacher-owned and audited.

Explainable by default



Screenshot: synthetic Patna NEET Academy · 400 students

DATA IN

From 1,205 messy rows to 1,200 usable responses.

A bad row becomes a visible exception — not a silent analytics failure.

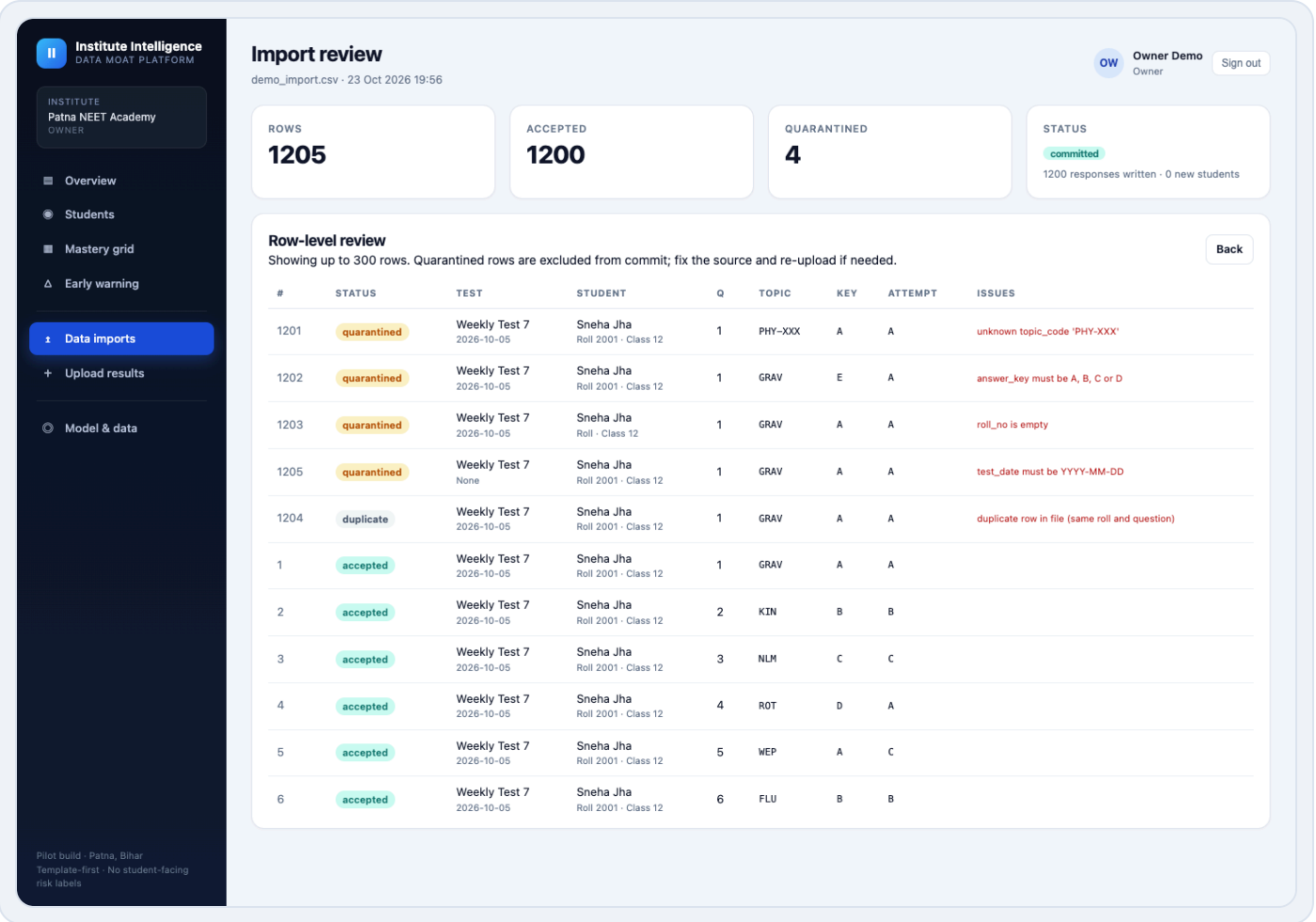
ROWS UPLOADED	ACCEPTED
1,205	1,200

QUARANTINED	DUPLICATE
4	1

WHAT THE PARSER CHECKS

Required columns · topic code · answer key · date format
question range · duplicate rows · test/batch consistency

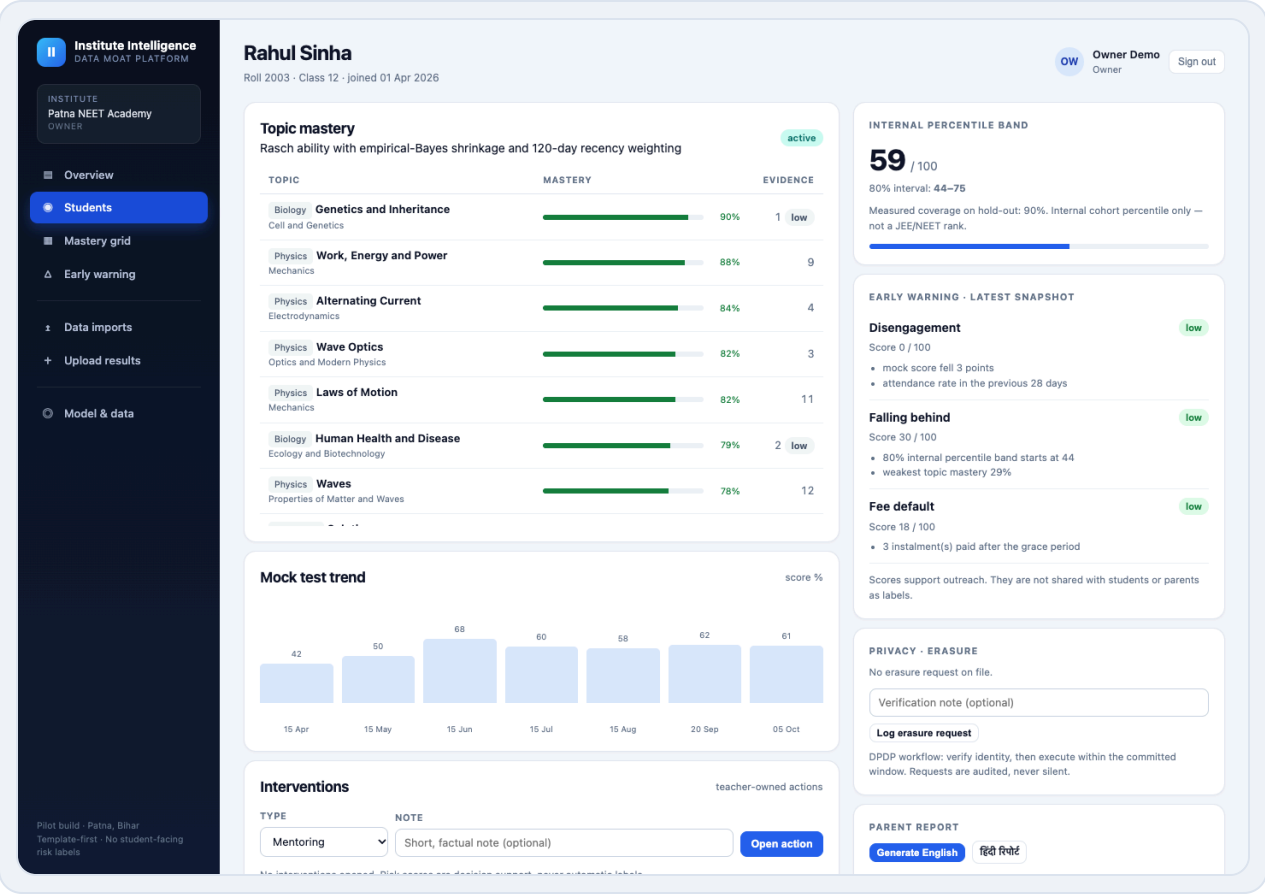
Review quarantined rows first. Commit only valid rows. Re-upload returns the same job.



TEACHER VIEW

A student view a teacher can act on.

Mastery, trend and risk reasons on one page — with the next step attached.



- **Mastery with evidence**
  Shrinkage + recency; low-evidence topics are flagged, not over-interpreted.
- **Risk reasons, not a black box**
  Attendance trend, score delta, fees where permitted.
- **Study plan**
  Three weakest topics retrieve curated concept and drill briefs.
- **Intervention**
  Mentoring, remedial, parent call or fee-plan review — audited.

No rank labels

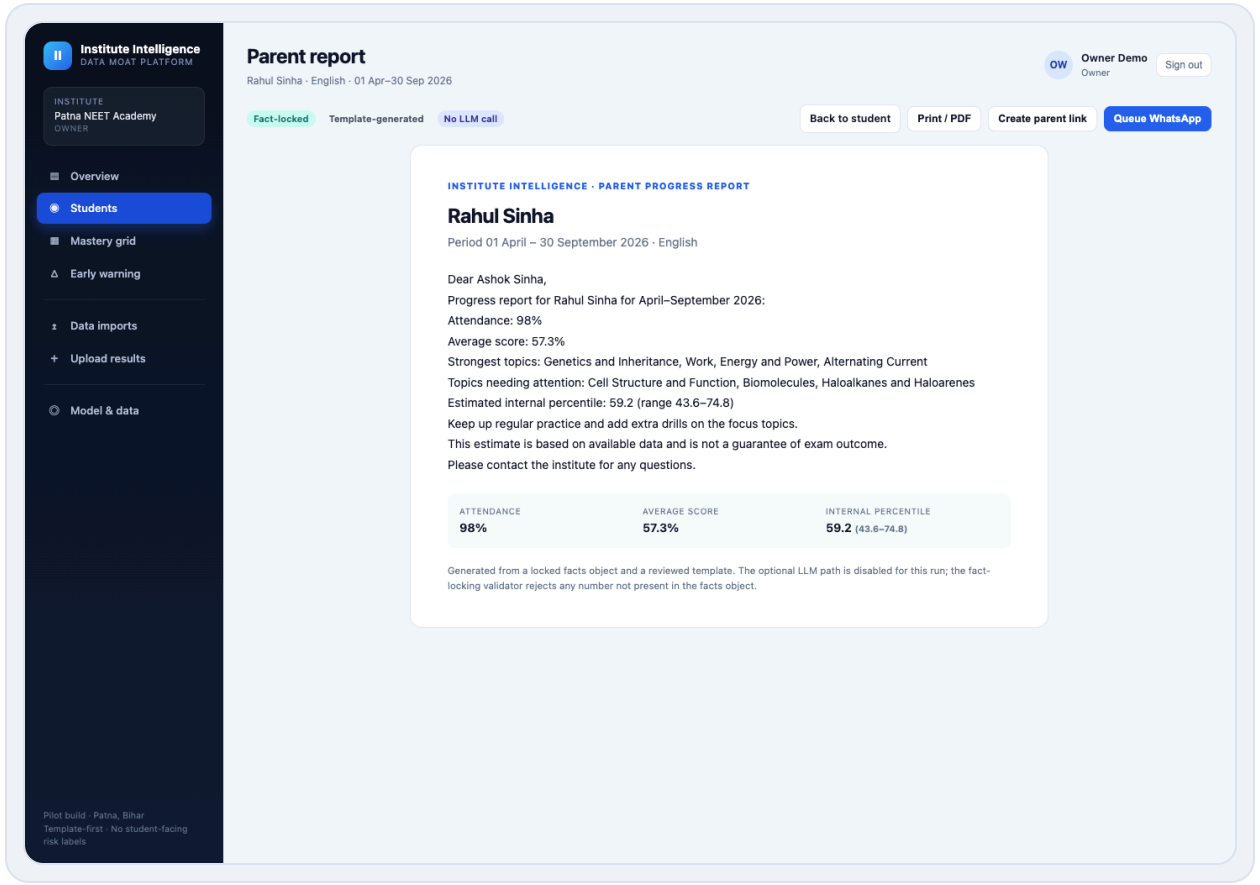
PARENT VIEW

A parent report that explains, not ranks.

Attendance, score, strong topics, focus topics and an internal percentile band.

- **Fact object**
  Every number in the report comes from the facts object. A validator rejects anything else.
- **English + Hindi**
  Same facts, reviewed template, deterministic fallback.
- **No risk. No fees.**
  Parents never see dropout labels or fee status.
- **90-day link**
  Token stored as SHA-256; consent required before creation.

Fact-locked · Bilingual · No exam rank



AI / ML DESIGN

AI where it earns its place.

Explainable statistics first. Learned models where the data supports them. Rules where it does not.

MEASURE

Rasch / 1PL

Ability and item difficulty from response patterns. Empirical-Bayes shrinkage (κ=8), 120-day half-life, 10 anchor items per test.

PREDICT

Calibrated disengagement

Logistic model on attendance, score and mastery features. 60/20/20 split, Platt scaling, top-10% review.

BE HONEST

Rules for thin outcomes

Fee default and falling behind are transparent v1 rules, labelled as rules in the model card.

COMMUNICATE

Locked facts + RAG-lite

Template-first report; every number validated. Study plans retrieve curated topic briefs. No student data to an external model.

No student-facing chatbot · No invented numbers · No black-box exam rank

GUARDRAIL
Model cards

GUARDRAIL
Fact-lock validator

GUARDRAIL
Rule vs model labels

GUARDRAIL
No external student data

VALIDATION

Validated on a 1,000-student synthetic pilot.

Enough to test the plumbing and the workflow. Not enough to claim real-world prediction.

Institute	AUC	PR-AUC	80% coverage	MAE
Patna NEET Academy	0.992	0.925	90.0%	8.07
Gaya Success Point	1.000	1.000	74.3%	8.38
Patna JEE Circle	0.922	0.738	84.0%	8.47

181,201 responses · Rasch converges in 41–76 iterations · 6/6 automated tests pass · rebuild 7.7 s

Read this before the AUC

Withdrawal is generated from attendance and fee behaviour. The model partly learns the generator. AUC 1.0 is not a real-world claim.

What this proves

Ingestion, fitting, calibration, reporting and privacy controls work end to end. The full analytics rebuild takes 7.7 s.

What it does not prove

That the model predicts real dropout. The next study is time-based, on consented pilot data, with a pre-registered success bar.

Pilot gate: 2 institutes · 8 weeks · 300–600 students · time-based holdout · pre-registered success bar.

PRIVACY BY ARCHITECTURE

Privacy is the architecture, not a policy page.

The parent view, the teacher view and the tenant boundary are different products.

INSTITUTE TENANT · RLS FORCED

student_pii

Names, DOB, guardian phone HMAC. Split from operational tables.

responses + snapshots

Item-level data, mastery, risk, percentile bands. Versioned.

consent + audit

Purpose-bound consent; every send, report and erasure logged.

job queue

SECURITY DEFINER functions; no direct app role access.

No tenant context → 0 rows returned.

CONTROLS IN THE PILOT

- **Tenant isolation**
  RLS forced on every tenant table; no-context reads return zero.
- **Role redaction**
  Teacher never sees fee amounts or fee-default risk. Parent never sees risk.
- **Consent gate**
  Report, link and WhatsApp blocked if parent communication consent is withdrawn.
- **Erasure workflow**
  Verified request, 30-day execution, audit trail.

PARENT VIEW

Attendance · score · topics · band

No risk. No fees. Token expires.

TEACHER VIEW

Risk + reasons · mastery · actions

Fee fields redacted server-side.

OWNER / ADMIN

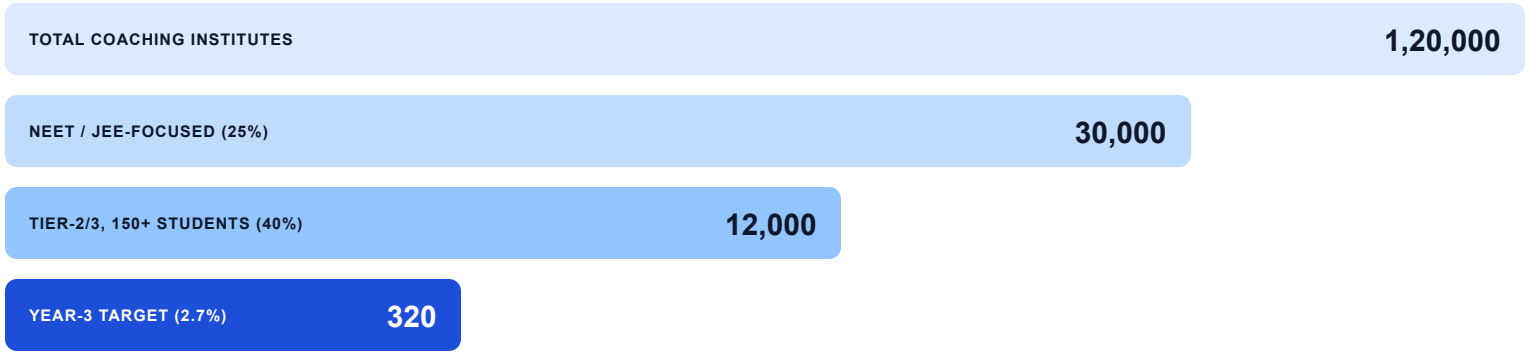
Fees · consent · imports · erasure

Full audit trail.

MARKET

A large, under-served market — with a narrow wedge.

Bottom-up, not a top-down TAM slide. Every number is an assumption to test.



WHY PATNA FIRST

Dense Tier-2 coaching cluster. Owner-led buying. Hindi + English parent communication. Short feedback loop for a design-partner cohort.

WHY NOW

WhatsApp is the parent channel

Updates are expected, but evidence is missing.

WHY NOW

Retention is the margin lever

A saved student is worth more than a new ad.

WHY NOW

DPDP raises the bar

Privacy becomes a buying criterion, not a footnote.

SAM = 12,000 institutes × blended ACV ₹1.0–1.1L ≈ ₹1,200–1,300 Cr

TAM/SAM are planning assumptions, not verified market research. Validate institute counts, willingness to pay and churn in the pilot.

BUSINESS MODEL

Unit economics work at ₹35–₹55 per student per month.

Annual contracts, 10 billed months, 150-student floor. Growth tier is the target mix.

GROWTH ACV
₹1.93L
350 students × ₹55 × 10 months

CONTRIBUTION
65.3%
After cloud, support, onboarding, messaging

CAC
₹38k
Founder-led sales + onboarding

PAYBACK
3.6 mo
Growth tier; 6.6 mo blended

LTV / CAC
4.6×
30-month conservative life

BREAK-EVEN
~110
Institutes at Year-1 fixed cost

Starter ₹35 · 200 students · ACV ₹70k | Growth ₹55 · 350 students · ACV ₹1.93L | 10 billed months | ₹1.0Cr ask

Service cost per institute: ₹25k Starter / ₹66.7k Growth. Fixed costs: ₹76L Year 1 → ₹1.2Cr Year 3.

YEAR-END ARR TRAJECTORY

YEAR 1 · 40 INSTITUTES
₹40L

YEAR 2 · 140 INSTITUTES
₹1.49Cr

YEAR 3 · 320 INSTITUTES
₹3.61Cr

GO-TO-MARKET

Wedge → proof → scale.

Sell the weekly workflow to owners. Prove retention and parent engagement. Expand by referral.

NOW

3 synthetic institutes

Working prototype; 181k responses; 6/6 tests.

NEXT 90 DAYS

10 design partners

2 paid pilots; 300–600 students; parent report study.

YEAR 1

40 institutes

≈9,500 students; native Forms; OMR adapter.

YEAR 2–3

140 → 320 institutes

34k → 80k students; UP/Jharkhand; benchmark network.

DESIGN PARTNER OFFER

Free 4-week historical audit

Prove value before workflow change.

PAID PILOT

₹25,000 for 8 weeks

Parent report study + teacher review.

CONVERT

Annual pre-paid contract

150-student floor; 10 billed months.

ASK · ₹1.0 CRORE / 18 MONTHS

Product 35% · pilots 20% · sales 20% · content/compliance 15% · buffer 10%

MILESTONES THAT UNLOCK THE NEXT ROUND

10 paying institutes · 60%+ parent report open rate · 3-point attendance lift · CAC payback under 9 months

Channel: owner referrals · local coaching associations · WhatsApp groups · one distributor per district. **Land:** 4-week historical-data pilot, then annual pre-paid.

THE MOAT

Start with 10 Patna institutes.
Build the trust layer for Indian coaching.

1,000
students in the synthetic pilot

181k
responses through the full pipeline

6/6
automated tests passing

The moat is the weekly workflow — and the permission to act on it.

DEMO LOGIN
owner@demo.test
demo1234